

the final frontier of this, might even be sensing brain signals from close to your head.” Choudhury and team are excited about the prospect of audible augmented reality headsets where signal processing technology can be used to map certain sounds onto your landscape. This could allow users to easily navigate their way through an airport, mall, museum, or any public space by walking towards



Debussy Prelude smart autonomous headphones

a sound that says something along the lines of “follow me” or “this way”. All of this without having to look down at your smartphone screens. Seems pretty cool.

Better batteries

Battery life of headphones has improved tremendously over the years, with manufacturers being able to offer impressive battery life despite headphones getting more compact. However, Lithium-ion batteries have their limitations and there’s a cap to how much we can expect out of them. The alternative is graphene batteries that have a higher energy density as compared to lithium-ion batteries. Lithium-ion stores up to 180Wh of energy per kilogram while graphene can store a whopping 1,000Wh per kilogram. We’ve already seen the use of graphene in headphone drivers where products that incorporate the material have

better sound and are comparatively more compact. Cell manufacturer VARTA has already developed graphene-enabled silicone batteries for wireless headphones, hearing aids, and other small electronic devices. The company strongly believes that silicone is the next-generation anode material in already present lithium-ion batteries. However, the company achieved much better results using graphene along with silicone. “Silicon increases the energy that can be stored in the battery, so we want to keep as much as possible – and thanks to graphene, we can do this without observing structural issues”, said Vittorio Pellegrini, from Graphene Flagship associated member BeDimensional. Graphene-powered batteries in headphones seem like the next logical progression to having much longer battery lives on wireless headphones, and we’re almost positive the technology will become omnipresent in the next 20 years.

New and improved technologies

A fair number of audio brands are working towards making headphones almost or completely independent, meaning they don’t require a source device to leverage all the features. These headphones are being dubbed as ‘streaming headphones’. The Debussy Prelude is an example of a streaming headset, in fact, the company calls the device a high-fidelity smart streamer. The headphones come with their own internal storage (32GB or 64GB depending on the variant) so that users can store tracks directly on the headphones and skip using the smartphone. These headphones also come with side-mounted displays, 4G and Wi-Fi connectivity, meaning you can stream songs from your favourite platforms, hence the nomenclature. The interface and capabilities of the device are fairly rudimentary at this point, but we see audio companies running with this idea in the future and coming

up with their own version of smart autonomous headphones.

We also predict that more and more headphones in the future will be contextually aware leading to headphones that are more proactive and not always user dependent for simple, repetitive tasks. Context aware headphones will employ various sensors and Artificial Intelligence to be aware of the user’s surroundings, current activity and more. We have already seen headphones capable of analysing the ambient sound and adjusting noise cancelling levels. In the future, we could see headphones capable of fitness tracking and fall monitoring based on your movements. We could also see the virtual assistants on headphones becoming more contextually-aware and suggesting playlists, podcasts, or more depending on our environment, time of day, and other factors. Wouldn’t it be nice to get on the train to work every morning and have your headphones automatically begin playing a podcast you’ve listened to at this particular time for a few days now?

Another interesting possibility for the future of headphones is direct sound beaming. Companies are working on sound beaming technology where music is directly ‘beamed’ to a person’s head without the use of any headphones or wires. An Israeli company, Noveto Systems, has already launched a device dubbed the SoundBeamer 1.0 that employs this exact idea. The device has a 3D sensing module that locates a person and then tracks the person’s ear’s position using ultrasonic waves to send audio directly to their ear. Sound pockets, of sorts, are created around the person’s ear and the user can listen to music, watch TV shows, and more without disturbing anyone or wearing bulky headphones. While the technology is still in its infancy and Noveto Systems is the only company to create a consumer item with the technology, we envision this technology becoming a lot more popular in the future, one that could possibly completely alter headphones as we know them. 